Experiment 1 : Multi-Page App using React Router

CODE:

Home.jsx :

```
import React from 'react';

const Home = () => {
  return (
    <div style={{ textAlign: 'center', padding: '20px' }}>
      <h2>Home Page</h2>
      <p>Welcome to our simple multi-page application.</p>
    </div>
  );
};

export default Home;
```

Contact.jsx :

```
import React from 'react';

const Contact = () => {
  return (
    <div style={{ textAlign: 'center', padding: '20px' }}>
      <h2>Contact</h2>
      <p>Email us at: support@example.com</p>
    </div>
  );
};

export default Contact;
```

About.jsx :

```
import React from 'react';

const About = () => {
  return (
    <div style={{ textAlign: 'center', padding: '20px' }}>
      <h2>About Us</h2>
      <p>This is a beginner-friendly app demonstrating React Router navigation.</p>
    </div>
  );
};

export default About;
```

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Navbar.jsx :

```
import React from 'react';
import { Link } from 'react-router-dom';

const Navbar = () => {
  return (
    <nav style={{ padding: '15px', backgroundColor: '#e0e0e0', textAlign: 'center', borderRadius:
'5px', margin: '20px' }}>
      <Link to="/" style={{ margin: '0 15px', textDecoration: 'none', color: '#000', fontWeight: 'bold'
}}>Home</Link>
      <Link to="/about" style={{ margin: '0 15px', textDecoration: 'none', color: '#000', fontWeight:
'bold' }}>About</Link>
      <Link to="/contact" style={{ margin: '0 15px', textDecoration: 'none', color: '#000', fontWeight:
'bold' }}>Contact</Link>
    </nav>
  );
};

export default Navbar;
```

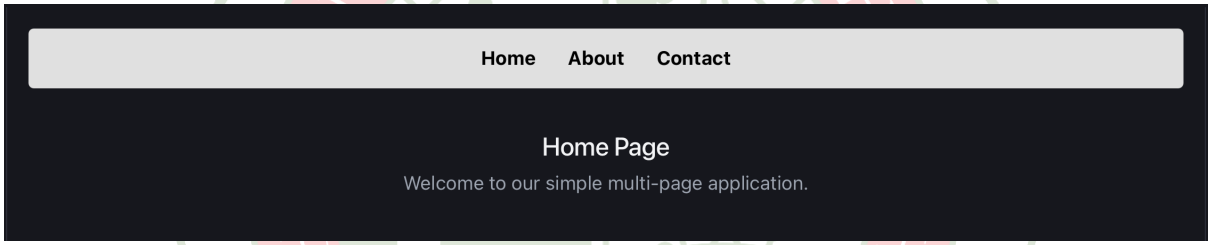
App.jsx :

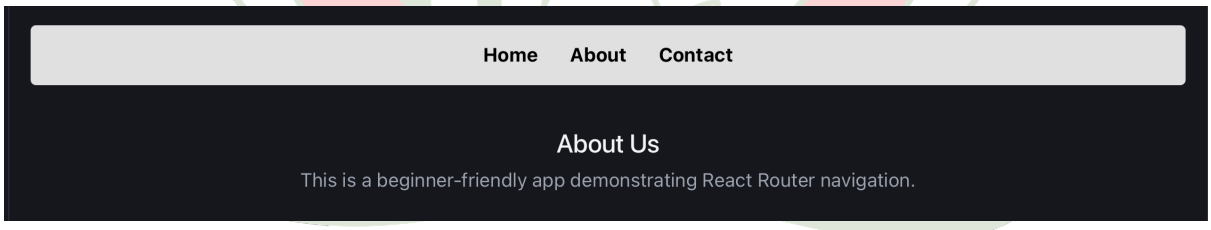
```
import React from 'react';
import { BrowserRouter, Routes, Route } from 'react-router-dom';
import Navbar from './Navbar';
import Home from './Home';
import About from './About';
import Contact from './Contact';

function App() {
  return (
    <BrowserRouter>
      <div>
        <Navbar />
        <Routes>
          <Route path="/" element={<Home />} />
          <Route path="/about" element={<About />} />
          <Route path="/contact" element={<Contact />} />
        </Routes>
      </div>
    </BrowserRouter>
  );
}

export default App;
```

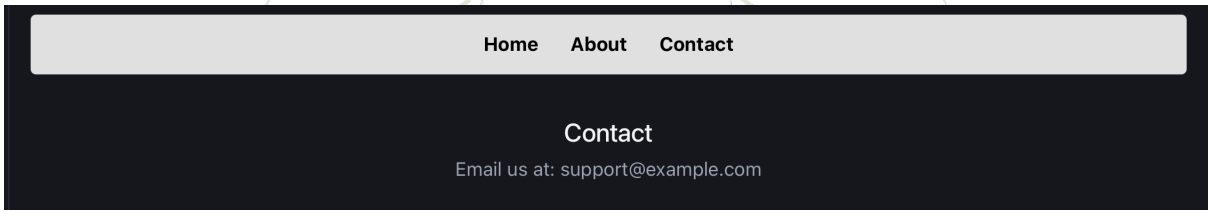
Main.jsx :

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <App />
  </StrictMode>,
)
```

OUTPUT:







Experiment 2: Student Table with Search & Pagination

CODE:

StudentTable.jsx :

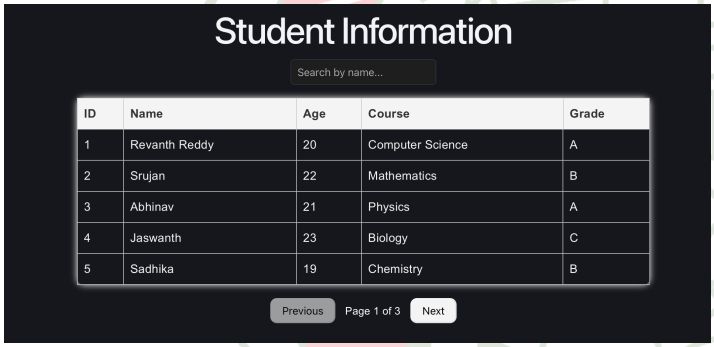
```
import React, { useState } from 'react';
import './StudentTable.css';

const initialStudents = [
  { id: 1, name: 'Revanth Reddy', age: 20, course: 'Computer Science', grade: 'A' },
  { id: 2, name: 'Srujan', age: 22, course: 'Mathematics', grade: 'B' },
  { id: 3, name: 'Abhinav', age: 21, course: 'Physics', grade: 'A' },
  { id: 4, name: 'Jaswanth', age: 23, course: 'Biology', grade: 'C' },
  { id: 5, name: 'Sadhika', age: 19, course: 'Chemistry', grade: 'B' },
  { id: 6, name: 'Potti', age: 20, course: 'Computer Science', grade: 'A' },
  { id: 7, name: 'VishnuTeja', age: 22, course: 'English', grade: 'B' },
  { id: 8, name: 'BalRaju', age: 21, course: 'History', grade: 'A' },
  { id: 9, name: 'Vamshi', age: 23, course: 'Mathematics', grade: 'C' },
  { id: 10, name: 'Annanya', age: 19, course: 'Physics', grade: 'B' },
  { id: 11, name: 'Samhitha', age: 20, course: 'Computer Science', grade: 'A' },
  { id: 12, name: 'Tejitha', age: 22, course: 'Biology', grade: 'A' },
  { id: 13, name: 'Ayush', age: 21, course: 'Chemistry', grade: 'B' },
  { id: 14, name: 'Saarthak', age: 23, course: 'History', grade: 'A' }
];

const StudentTable = () => {
  const [searchQuery, setSearchQuery] = useState('');
  const [currentPage, setCurrentPage] = useState(1);
  const recordsPerPage = 5;

  const handleSearch = (e) => {
    setSearchQuery(e.target.value);
    setCurrentPage(1); // Reset to first page when search changes
  };

  const filteredStudents = initialStudents.filter(student =>
    student.name.toLowerCase().includes(searchQuery.toLowerCase())
  );

  const totalPages = Math.ceil(filteredStudents.length / recordsPerPage);

  const indexOfLastRecord = currentPage * recordsPerPage;
  const indexOfFirstRecord = indexOfLastRecord - recordsPerPage;
  const currentRecords = filteredStudents.slice(indexOfFirstRecord, indexOfLastRecord);

  const prevPage = () => {
    if (currentPage > 1) setCurrentPage(currentPage - 1);
  };

  const nextPage = () => {
    if (currentPage < totalPages) setCurrentPage(currentPage + 1);
  };
```

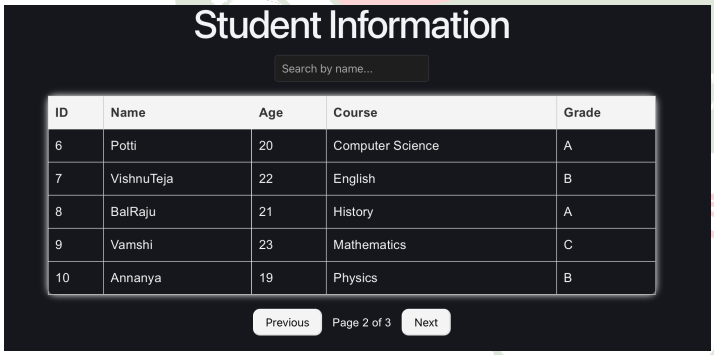
```
  return (
    <div className="simple-container">
      <div>
        <input
          type="text"
          placeholder="Search by name..."
          value={searchQuery}
          onChange={handleSearch}
        />
      </div>

      <table border="1" cellPadding="10" className="simple-table">
        <thead>
          <tr>
            <th>ID</th>
            <th>Name</th>
            <th>Age</th>
            <th>Course</th>
            <th>Grade</th>
          </tr>
        </thead>
        <tbody>
          {currentRecords.length > 0 ? (
            currentRecords.map(student => (
              <tr key={student.id}>
                <td>{student.id}</td>
                <td>{student.name}</td>
                <td>{student.age}</td>
                <td>{student.course}</td>
                <td>{student.grade}</td>
              </tr>
            ))
          ) : (
            <tr>
              <td colSpan="5">No students found.</td>
            </tr>
          )}
        </tbody>
      </table>

      <div>
        <button onClick={prevPage} disabled={currentPage === 1 || totalPages === 0}>
          Previous
        </button>
        <span className="page-text">
          Page {currentPage} of {totalPages || 1}
        </span>
        <button onClick={nextPage} disabled={currentPage === totalPages || totalPages === 0}>
          Next
        </button>
      </div>
    </div>
  );
};

export default StudentTable;
```

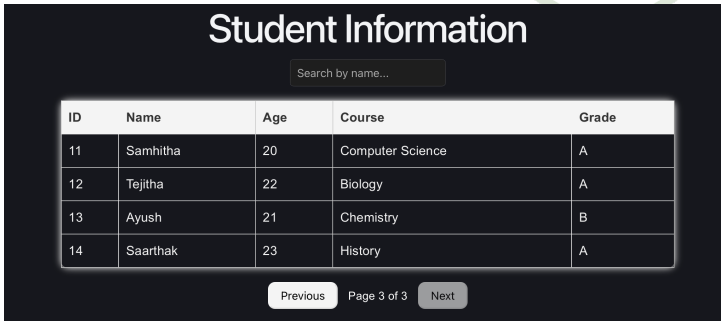
App.jsx :

```
import React from 'react';
import StudentTable from './StudentTable';
import './App.css';

function App() {
  return (
    <div className="App">
      <h1>Student Information</h1>
      <StudentTable />
    </div>
  );
}

export default App;
```

OUTPUT:







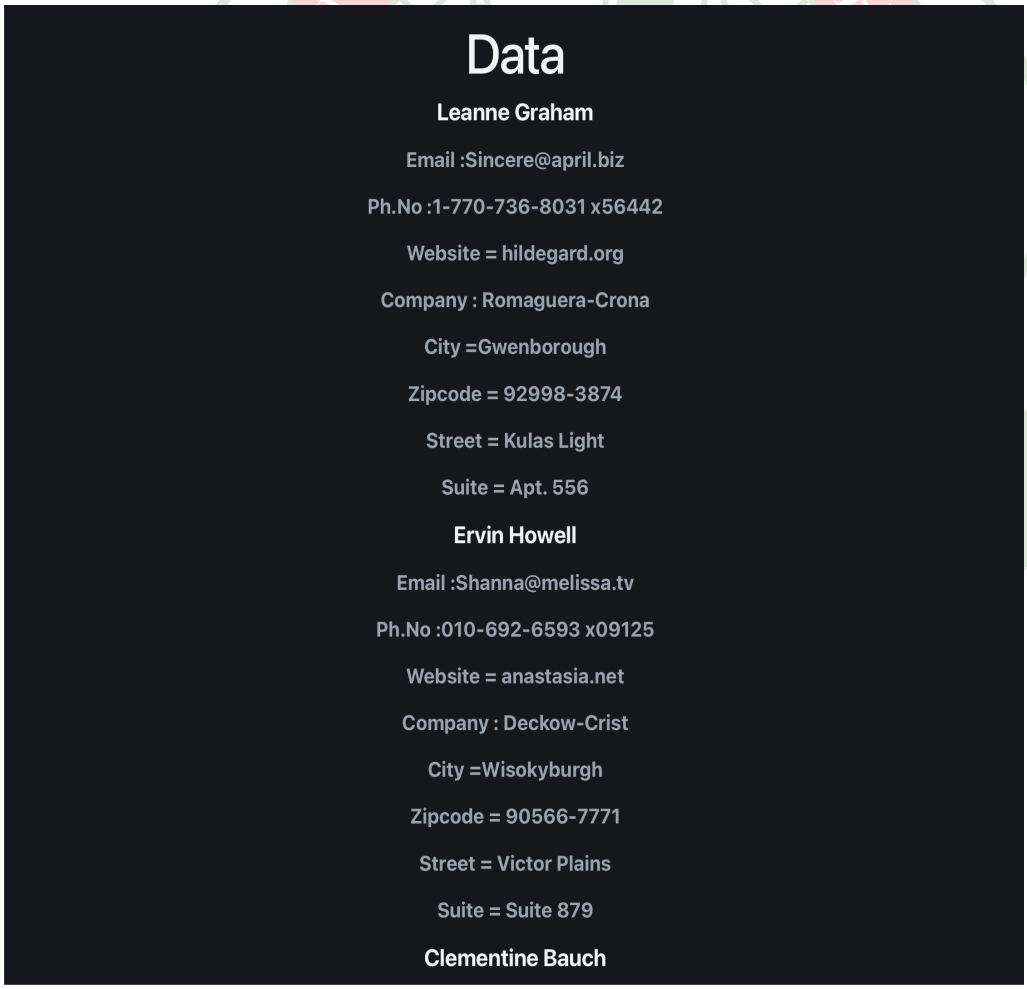
Experiment 3 : Api Calls

CODE:

GetData.jsx :

```
import React from "react";
import { useState, useEffect } from "react";

function GetData() {
   const URL = "https://jsonplaceholder.typicode.com/users"

   const [printData, setPrintData] = useState([]);

  const apiCalls = async () => {
   const response = await fetch(URL);
   console.log(response)
   const data = await response.json();
   console.log(data)
   setPrintData(data);
  };

  useEffect(() => {
   apiCalls();
  }, []);

  return (
   <div>
      <h1>Data</h1>
      {
         printData.map(function(item){
            return(
               <div key={item.id}>
                  <h2 style={{fontWeight:"bold"}}>{item.name}</h2>
                  <h3>Email :{item.email}</h3>
                  <h3>Ph.No :{item.phone}</h3>
                  <h3> Website = {item.website}</h3>
                  <h3>Company : {item.company.name}</h3>
                  <h3>City ={item.address.city}</h3>
                  <h3>Zipcode = {item.address.zipcode}</h3>
                  <h3>Street = {item.address.street}</h3>
                  <h3>Suite = {item.address.suite}</h3>

               </div>
            )
         })
      }
   </div>
  );
}
export default GetData;
```

App.jsx :

```
import react from "react";
import GetData from "./GetData";

function App() {
  return (
    <>
      <GetData />
    </>
  );
}

export default App;
```

OUTPUT: